

23	A	Consider the secondary circuit: when temperature is low the resistance of the thermistor is high. This means that the terminal p.d. across the secondary cell is larger. (note that if the secondary cell has no internal resistance it would not matter what the resistance of the thermistor was, as the terminal p.d. would always simply be the secondary cell's e.m.f.) Consider the driver circuit: when it is dark the resistance of the LDR increases. This means that less of the driver cell e.m.f. drops across wire PQ. With a smaller p.d. per unit length of PQ, a longer balance length is needed.
24	B	For the first part, the magnetic field is $B = \frac{\mu_0 I}{2\pi r}$ and the force on the wire is $F = BIL = \frac{\mu_0 I^2 L}{2\pi r}$.